



The longitudinal associations among student externalizing behaviors, teacher-student relationships, and classroom engagement

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ABSTRACT

Personal characteristics and classroom environment features both play important roles in predicting students' levels of classroom engagement. The present study took a person-environment transaction perspective to investigate how factors at both the personal (i.e., student externalizing behaviors) and relational (i.e., teacher-student relationships) levels jointly predict the development of classroom engagement behaviors in a sample of 784 elementary school students. Using a longitudinal cross-lagged model spanning across Grade 3 to Grade 5, we found a negative reciprocal association between teacher-student relationships and externalizing behaviors, such that a more positive teacher-student relationship predicted fewer externalizing behaviors in the subsequent academic year, and fewer externalizing behaviors predicted a more positive teacher-student relationship 1 year later. In addition, externalizing behaviors directly negatively predicted subsequent classroom engagement, whereas teacher-student relationships indirectly predicted subsequent classroom engagement by way of externalizing behaviors. Overall, students with more externalizing behaviors experienced more conflicts with and received less support from their teachers, which predicted the development of more externalizing behaviors and lower subsequent classroom engagement.

Positive and sustained engagement in the classroom is essential for quality learning, whereas low engagement is a primary driving force for poor achievement outcomes (Fredricks et al., 2016; Prince, 2004; Willson & Hughes, 2009). Elementary school is when students are expected to master knowledge and skills foundational for advanced learning in subsequent educational stages. As such, the extent to which students engage in classroom learning in elementary school consistently and reliably predicts students' long-term academic outcomes, such that a higher level of early engagement predicts higher academic achievement as well as a lower likelihood of dropout in high school (Alexander et al., 2001; Lei et al., 2018; Li et al., 2022). Unfortunately, the average classroom engagement begins to decline as early as late elementary school (Wylie & Hodgen, 2012). Given the long-term cascading effects of

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early classroom engagement on subsequent learning outcomes, it is important to understand factors that promote versus those that undermine students' classroom engagement in elementary school.

Previous studies have shown that personal characteristics and relational quality in the classroom environment both play important roles in predicting students' levels of classroom engagement (e.g., Hughes et al., 2008; Olivier, Morin, et al., 2020; Shin & Chang, 2022). Although the existing literature includes numerous works that have investigated how personal level factors and relational level factors separately contribute to classroom engagement, few have considered the person-environment transactional processes, including how the joint effects of these personal and relational level factors may shape students' classroom engagement (Archambault et al., 2017). It is possible that students' personal characteristics may indirectly influence their engagement by shaping their social relationships within the classroom. Similarly, students' social relationships may also indirectly influence their engagement by promoting or undermining students' capacity to engage in the classroom. By considering only one factor at a time while ignoring these critical person-environment transactions, researchers and educators miss the opportunity to understand students' engagement as a developing process evolving from the dynamic interaction between students and their social environments. To address this major gap in the literature, the present study used a multi-informant approach in a 3-year longitudinal design and utilized a person-environment transactional perspective to investigate the joint roles of students' externalizing behaviors (i.e., a personal level factor) and teacher-student relationships (i.e., a relational factor) in the development of classroom engagement in a sample of elementary school students. Fig. 1 presents the conceptual model that guided the literature review and development of hypotheses.

1. Teacher-student relationships and classroom engagement

Classroom engagement is a multi-faceted construct, encompassing engagement at the behavioral, cognitive, emotional, and social levels (Fredricks et al., 2004; Wang et al., 2016). Engagement at the behavioral level, defined as active participation and involvement in class activities, is the strongest and most robust predictor of academic achievement among all engagement dimensions (Fredricks et al., 2004; Lei et al., 2018; Wang et al., 2016). Thus, the present study operationalized classroom engagement at the behavioral level.

A conducive learning environment is imperative for fostering students' motivation to engage in learning (Fig. 1 Paths A; Patrick et al., 2007). A conducive learning environment is first characterized by high quality classroom instruction such as practices to engage students in classroom discourse and active discussion, encourage deep-level processing, and promote mastery-oriented learning goals (Church et al., 2001; Patrick et al., 2007; Stipek & Chiatovich, 2017). In addition to instructional practices, another important aspect that contributes to the establishment of a conducive classroom environment is a positive social environment shaped, in part, by a positive teacher-student relationship. A teacher-student relationship is consistently defined by two moderately correlated dimensions: teacher support and teacher-student conflict (Pianta, 1999; Roorda et al., 2017; Wu & Hughes, 2015). In a warm and supportive teacher-student relationship, the interactions between the teacher and the students are based on encouragement, trust, and respect (Furrer et al., 2014), which all help foster a sense of relatedness and belongingness (Allen et al., 2018; Chiu et al., 2016). According to Self-Determination Theory, a sense of emotional security allows students to fully engage in learning that is centrally important for fostering students' intrinsic motivation and internalizing the regulation for positive school-related values and behaviors (Ryan & Deci, 2000). Indeed, the positive relational features of the classroom environment, such as high levels of warmth, support, and positive

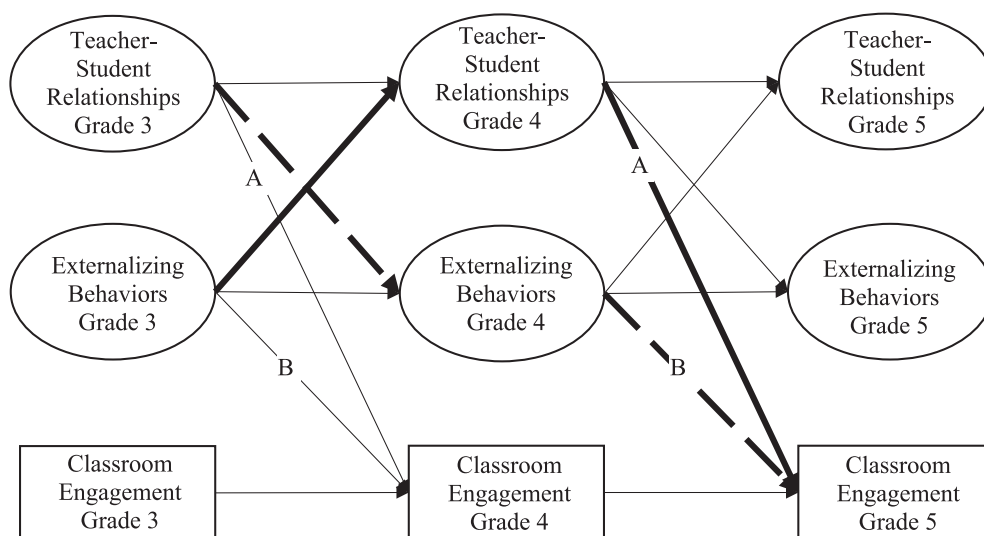


Fig. 1. Conceptual Model.

Note. Paths A represent the longitudinal association between teacher-student relationships and classroom engagement; Paths B represent the longitudinal association between externalizing behaviors and classroom engagement; bolded solid paths represent the longitudinal association between teacher-student relationships and classroom engagement mediated by externalizing problems; bolded dashed paths represent the longitudinal association between externalizing behaviors and classroom engagement mediated by teacher-student relationships.

affective characteristics, are evidenced to motivate students to engage in learning within the classroom (Hamre & Pianta, 2001; Hughes & Kwok, 2006; Klem & Connell, 2004; Quin, 2017; Roorda et al., 2017; Shin & Chang, 2022). In contrast, a conflictual teacher-student relationship is characterized by discordant interactions in which teachers may attempt to frequently control and harshly respond to students' behaviors, further resulting in students' anger and anxiety and subsequent withdrawal from school learning activities (Birch & Ladd, 1997; Hamre & Pianta, 2001; Olivier, Morin, et al., 2020). Importantly, teacher-student relationship quality may be more impactful in shaping students' classroom engagement in elementary school than in other educational stages because students are decreasingly adult-oriented (e.g., teacher) and increasingly peer-oriented (e.g., classmates) as they transition from elementary to middle school (Lynch & Cicchetti, 1997). Consistent with this developmental perspective, Roorda et al.'s (2011) meta-analysis indicated that the conflict dimension of teacher-student relationship is associated with students' classroom engagement more strongly in elementary school than in middle school.

2. Externalizing behaviors and classroom engagement

The extent to which children engage in learning in the classroom also varies as a function of their capacity to do so. For example, children with more externalizing behaviors are found to be less capable of engaging in learning in the classroom setting (Fig. 1 Paths B; Olivier, Archambault, & Dupere, 2020; Olivier, Morin, et al., 2020; Searle et al., 2013; Tian et al., 2017; Wang & Fredricks, 2014). Children's externalizing behaviors are characterized by hyperactivity, noncompliance, aggression, and rule-breaking (Achenbach & Edelbrock, 1978; Keil & Price, 2006). These are outward-directed behaviors that negatively impact their external environments and social encounters, such as their parents, peers, and teachers, which are highly incompatible with the academic and behavioral expectations of the classroom. Students with higher externalizing behaviors experience more difficulties in regulating their attention, emotion, and behaviors (Rothbart & Bates, 2006) and are less able to actively engage in learning by paying attention to teachers' instructions, persisting on tasks, and working collaboratively with classmates (Baker et al., 2008; Campbell et al., 2014).

3. Person-environment transaction and classroom engagement

Although teacher-student relationships are often conceptualized as the environmental context that either fosters or suppresses students' engagement in the classroom (Furrer et al., 2014; Klem & Connell, 2004; Quin, 2017; Roorda et al., 2017), it is important to acknowledge the person-environment transactional processes in the formation and maintenance of these relationships (Hamre & Pianta, 2001; Hughes & Kwok, 2006) as well as in the subsequent development of classroom engagement behaviors. Specifically, a teacher-student relationship is the result of an ongoing interaction between a developing student and their changing contexts (Sameroff, 2009). Individuals play an active role in creating their social environments, such that the unique characteristics of each person tend to elicit certain patterned responses from their social counterparts (Sameroff, 2009; Scarr & McCartney, 1983). Children with more externalizing behaviors are consistently shown to evoke harsher and more negative responses in their social interactions (Fontaine et al., 2002). For example, as externalizing behaviors are highly disruptive to the classroom environment, a teacher may attempt to restore the classroom order by responding to these externalizing behaviors in a negative controlling manner, thus escalating conflicts in the teacher-student dyad (Buyse et al., 2008; Hendrickx et al., 2022; Leflot et al., 2011). Such a negative teacher-student relationship is shown to result in a lack of sense of relatedness and belongingness in school, which are critical social motivations for the development of classroom engagement (Olivier, Archambault, & Dupere, 2020; Olivier, Morin, et al., 2020). In this way, children's externalizing behaviors may indirectly influence their classroom engagement by way of shaping their social relationships and social motivation for engagement (see Fig. 1 bolded solid paths). However, the proposition that children's externalizing behaviors may negatively predict their classroom engagement by way of shaping their social relationships has not been investigated.

Additionally, numerous studies have demonstrated that teacher-student relationships reciprocally influence the development of externalizing behaviors (Baker et al., 2008; Harvey et al., 2022; Leflot et al., 2011; O'Connor et al., 2011; Silver et al., 2005; Williford et al., 2017). The ways in which teachers respond to externalizing behaviors shape the affective climate in the classroom and model acceptable behaviors in social interactions. A warm and supportive teacher-student relationship promotes students' trust and feelings of security toward the teacher as well as models a nonreactive, well-regulated response style, which mitigates the development of externalizing behaviors (Baker et al., 2008; Leflot et al., 2011; O'Connor et al., 2011; Silver et al., 2005; Williford et al., 2017). Conversely, a conflictual teacher-student relationship fosters a sense of insecurity and mistrust and models a reactive and aggressive response style, which exacerbates the development of externalizing behaviors (Baker et al., 2008; Leflot et al., 2011; O'Connor et al., 2011; Silver et al., 2005; Williford et al., 2017). As such, it is possible that a high-quality positive teacher-student relationship may also indirectly promote students' classroom engagement by lessening their externalizing behavioral symptoms such as disruptiveness and inattentiveness (see Fig. 1 bolded dashed paths). This possible indirect effect of teacher-student relationships on the development of classroom engagement by way of shaping students' externalizing behaviors has not been examined in the existing literature and requires further investigation.

A review of the literature reveals only one existing study that considered both externalizing behaviors and teacher-student relationships simultaneously in relation to classroom engagement (Archambault et al., 2017). Using a sample of third through fourth grade elementary school students, Archambault and colleagues reported that more oppositional behaviors in the fall semester predicted lower classroom engagement in the following spring semester, an effect that was moderated by teacher-student relationships. Although this study demonstrated that a positive teacher-student relationship mitigated the negative effect of oppositional behaviors on classroom engagement, it did not discuss the likelihood that a highly oppositional student would experience such positive teacher-student relationships. As reported in Archambault et al., students' oppositional behaviors were associated moderately positively with

teacher-student conflict ($r = 0.32$), suggesting that highly oppositional students were less likely to enjoy positive teacher-student relationships than those with low oppositional behaviors. As a result, Archambault et al.'s study design did not allow for the evaluation of the effect of students' externalizing behaviors on their engagement behaviors situated within an ever-evolving social relational reality unique to each developing individual. The present study aimed to address this important gap by investigating the long-term development of students' classroom engagement in the context of these critical person-environment transactional processes.

4. The present study

Although levels of classroom engagement can vary as a function of both personal characteristics (e.g., externalizing behaviors) and relational features (e.g., teacher-student relationships), prior studies have focused on separately examining these factors (Olivier, Morin, et al., 2020; Quin, 2017; Roorda et al., 2017; Wang & Fredricks, 2014). The critical issue with this type of design is the failure to consider the transactional nature of the person-environment relation in the development of students' classroom engagement. Students are not randomly assigned into teacher-student relationships of varying qualities; rather, the unique relationship between each student and their teacher is partly shaped by the student's attributes. Similarly, students' externalizing behaviors are not developmentally fixed; rather, they may improve or deteriorate over time depending on the quality of their social interactions (e.g., amount of relational support and conflict). By considering only one factor at a time while ignoring these critical person-environment transactions, researchers and educators are left with an incomplete understanding of the full cascading process leading to the development of varying levels of classroom engagement. Consequently, when a student disengages in the classroom, we may fail to identify the true cause for their lack of engagement, falsely attribute their lack of engagement to one party, and miss the opportunity to intervene effectively.

To address this major gap in the literature, the present study investigated how students' externalizing behaviors and relationships with their teachers jointly predicted the development of their engagement behaviors in elementary school. We first examined whether there was a transactional relation between students' externalizing behaviors and their relationships with teachers as indicated by the existing literature (Lei et al., 2016; MacLean et al., 2023). Next, we examined whether students' externalizing behaviors and their relationships with teachers each directly and uniquely predicted the development of classroom engagement after controlling for the effects of the other predictor. Finally, we examined whether students' externalizing behaviors and their relationships with teachers each indirectly predicted the development of classroom engagement through the other predictor.

Using a multi-informant approach (i.e., student, teacher, and parent reports) in a multi-wave longitudinal design, our goal was to obtain a more holistic understanding of how these relations unfold over the course of elementary school years. We developed three hypotheses. First, based on the existing literature on the association between students' externalizing behaviors and teacher-student relationships (Crockett et al., 2018; Doumen et al., 2008; Harvey et al., 2022; MacLean et al., 2023; O'Connor et al., 2011; Silver et al., 2005; Williford et al., 2017; Zhang & Sun, 2011), we hypothesized that there would be a negative reciprocal relation between students' externalizing behaviors and positive teacher-student relationships. Second, based on the person-environment transactional perspective (Sameroff, 2009; Scarr & McCartney, 1983), as well as the literature on the association between students' externalizing behaviors and classroom engagement (Olivier, Archambault, & Dupere, 2020; Searle et al., 2013) and between teacher-student relationships and classroom engagement (Quin, 2017; Roorda et al., 2017; Shin & Chang, 2022), we hypothesized that externalizing behaviors would negatively predict subsequent student classroom engagement both directly and indirectly through teacher-student relationships. Similarly, we hypothesized that teacher-student relationships would positively predict subsequent student classroom engagement both directly and indirectly through student externalizing behaviors.

5. Method

5.1. Participants

Participants were part of the Impact of Grade Retention: A Development Approach Study (Hughes, 2017), a longitudinal study that collected data from 784 students in three public school districts in Texas (one in an urban district and two in a small city). Data were collected annually from these students, their parents, and teachers over the course of 14 years starting from Grade 1 (in 2001–2002). The present study analyzed data collected when students were in Grade 3 (in 2003–2005) through Grade 5 (in 2005–2007). Of the 784 students recruited, 372 were females and 412 were males (37.4% Hispanic, 34.1% White, 23.2% African American, 3.6% Asian/Pacific Islander, and 0.3% Native American/Alaskan). The percentage of students who had higher Spanish language proficiency than English proficiency were 11% in Grade 3, 9% in Grade 4, and 4% in Grades 5. In total, 80% of students scored higher on English language proficiency across Grades 3–5. Of the parents who participated, 9% reported receiving elementary/middle school education; 34% had either a high school degree, a GED or attended a vocational or trade school; 31% had some college education or an associate degree; 15% had a bachelor's degree; and 11% had a graduate degree. Approximately 67% of the teachers were female. Data collected when the students were in the third, fourth, and fifth grades were included in the current analyses when students' externalizing behaviors, teacher-student relationships, and classroom engagement were assessed. The present study analyzed de-identified secondary data, which is not considered Human Subject Research at the authors' institutions.

5.2. Procedure

During each academic year, students were interviewed and tested in an individual setting. Parents and teachers were mailed questionnaire packets to fill out. Interviews and surveys were conducted in late Fall or the Spring of an academic year to allow raters to

develop adequate knowledge of the child. Cognitive/achievement assessments were conducted throughout the year, with annual assessments being separated by approximately 12 months. The student cognitive/achievement assessments were conducted by research staff members. Prior to conducting assessments in the school, all research staff members (a) received a minimum of 12 h of instruction in the assessment to be conducted and (b) demonstrated a high proficiency in the test administration in practice sessions. In addition, all test protocols were checked for errors by members of the research staff and assessors were required to correct the error in the few cases in which errors were found.

5.3. Measures

The following criteria were used to evaluate the psychometric properties of the measures. Cronbach's alpha was used to evaluate internal consistency reliability. Given that Cronbach's alpha varies as a function of instrument length, a value of 0.70 is considered acceptable for scales with 10 or fewer items and 0.80 is considered acceptable for scales with >10 items (Cortina, 1993). The inter-rater agreement was evaluated based on the benchmark provided in Achenbach et al. (1987), with an acceptable inter-rater agreement being $r = 0.28$ or above. The criterion validity for a construct was evaluated based on the average correlation between the construct and relevant criterion variables in the existing literature (often reported in relevant meta-analyses).

5.3.1. Language proficiency test

The Woodcock-Muñoz Language test (Woodcock & Muñoz-Sandoval, 1993) was administered to students who were bilingual or primarily spoke Spanish; this assessment determined students' oral language, reading, and writing proficiency in English and Spanish. Reported internal reliability for both the broad English ability and broad Spanish ability were consistently above 0.90 (Woodcock & Muñoz-Sandoval, 1993). If a student demonstrated more proficiency in Spanish than English, all measures were administered in Spanish. After a student demonstrated equal or greater language proficiency in English than in Spanish for 3 consecutive years, all subsequent tests were administered in English. For students who spoke any Spanish or who were enrolled in bilingual education, both English and Spanish versions of the questionnaires were mailed to their parents.

5.3.2. Classroom engagement

Teachers reported students' classroom engagement using an adapted version of the Teacher and Student Ratings of the Classroom Engagement Scales (Skinner et al., 1998). Eleven of the 18 original items were retained to measure classroom engagement because the remaining seven items did not load well onto the classroom engagement construct in the current sample (Hughes, 2017). Teachers reported students' classroom engagement levels using a 4-point Likert scale (1 = *not at all true*, 4 = *very true*). Sample items included "When this student is in class, he/she concentrates on doing his/her work" and "When this student is in class, he/she participates in class discussion". Higher scale scores indicate more engagement in the classroom. This scale had acceptable internal reliability in the present sample, with Cronbach's alpha ranging from 0.93 to 0.94 in Grades 3–5. One-year stability of classroom engagement ranged from 0.52 to 0.55 in Grades 3–5. This adapted scale demonstrated acceptable criterion validity and was modestly positively correlated with students' standardized academic achievement scores ($r = 0.26$ – 0.27), which aligned with the correlation between engagement and achievement reported in Lei et al.'s (2018) meta-analysis (weighted $r = 0.27$). Several published studies reported acceptable psychometric properties for the original scale in other samples, including acceptable internal consistency reliability (Cronbach's alpha = 0.87 – 0.88), inter-rater agreement between teacher and student reports ($r = 0.32$ – 0.37), and criterion validity as demonstrated by its association with students' academic achievement ($r = 0.40$ – 0.58 ; Skinner et al., 1990, 2009).

5.3.3. Externalizing behaviors

Children's externalizing behaviors were assessed using two scales of the Strengths and Difficulties Questionnaire (SDQ; Goodman, 2001). Parents reported their child's externalizing behaviors using the conduct problems and hyperactivity scales, both of which contained five items. An example item for the conduct problem scale is "Generally well behaved, usually does what adults request" (reverse scored) and an example item for the hyperactivity scale is "Restless, overactive, cannot stay still for long". Parents provided their ratings on a 3-point Likert scale (0 = *not true*, 1 = *somewhat true*, and 2 = *certainly true*). Higher scale scores indicate higher levels of conduct problems and hyperactivity. In Grades 3–5, Cronbach's alpha ranged from 0.69 to 0.73 for the conduct problem scale and from 0.79 to 0.81 for the hyperactivity scale. One-year stability was 0.66 for the conduct problems subscale and from 0.72 to 0.77 for the hyperactivity subscale. Acceptable inter-rater reliability was evident for both subscales as they were correlated positively with the corresponding teacher reported subscale scores in Grades 3 and 4 ($r = 0.29$ – 0.36 for conduct problems; $r = 0.38$ – 0.45 for hyperactivity).¹ Stone et al.'s (2010) meta-analysis reported evidence of acceptable psychometric properties of the SDQ and its subscales in elementary school students. The weighted average Cronbach's alphas were 0.58 for conduct problems and 0.76 for hyperactivity. Although the average internal consistency was low for conduct problems, the meta-analysis reported a wide range of internal consistencies, with some studies showing quite low internal consistency (0.46; Parkes et al., 2008) and others reporting acceptable Cronbach's alpha (0.76; Janssens & Deboutte, 2009). Stone et al. (2010) also reported acceptable inter-rater agreement between parent and teacher reports ($r = 0.34$ for conduct problems; $r = 0.47$ for hyperactivity) and acceptable criterion validity as indicated by the associations between the SDQ and the corresponding scales in the Child Behavior Checklist ($r = 0.71$ for conduct problems; $r = 0.69$ for

² Teacher reported conduct problems and hyperactivity subscales in Grade 5 had unsalvageable coding errors, making them unusable for the present analyses.

hyperactivity; Achenbach, 1991; Stone et al., 2010).

5.3.4. Teacher-student relationships

Teacher-student relationships were measured using both students' and teachers' reports. Students used the Network of Relationship Inventory (NRI; Furman & Buhrmester, 1985) and were asked to rate the level of support (16 items) that their teachers provided and the level of conflict (six items) they had with their teachers. An example item for the support subscale is "How much does your teacher like or approve of the things you do?" and an example item for the conflict subscale is "How much do you and your teacher disagree and quarrel?". Students used a 5-point scale (1 = *little or none*, 5 = *the most*) to respond to each item.

Teachers used the Teacher Network Relationship Inventory (TNRI; Hughes et al., 2008), which is an adapted version of the NRI, and were asked to use a 5-point Likert scale (1 = *not true at all* to 5 = *very true*) to rate the level of support (16 items) and the level of conflict (six items) in their relationship with the target student. An example item for the support subscale was "This child accepts my help with things he/she cannot do by himself/herself" and an example item for the conflict subscale was "This child and I often argue with each other". Higher scores on the support subscale indicate more support toward the target student and higher scores on the conflict subscale indicate more conflict with the target student.

In several existing studies conducted among elementary school students, the NRI has demonstrated acceptable psychometric properties. For example, the NRI has been shown to have acceptable internal consistency reliability (Cronbach's $\alpha = 0.80\text{--}0.94$; Furman & Buhrmester, 1985; Hughes et al., 1997). In addition, teacher-student relationship conflict measured by the NRI has been associated with students' aggressive behaviors ($r = 0.26$; Hughes et al., 1997), which aligns with meta-analytic results examining the associations between teacher-student relationships and various types of student aggression ($r = 0.19$ to 0.55 ; Krause & Smith, 2022, 2023; Lei et al., 2016), indicating acceptable criterion validity for the NRI scale. In the present study, both student- and teacher-reported support and conflict subscales had acceptable internal reliability, with Cronbach's α ranging from 0.77 to 0.82 for student-reported support, 0.78 to 0.82 for student-reported conflict, 0.94 to 0.95 for teacher-reported support, and 0.91 to 0.93 for the teacher-reported conflict scale. One-year stability ranged from 0.37 to 0.44 for student-reported support, 0.39 to 0.46 for student-reported conflict, 0.31 to 0.37 for teacher-reported support, and 0.55 to 0.61 for teacher-reported conflict. Inter-rater agreement for the teacher- and student-reported conflict subscales were 0.39–0.42. Teacher-reported support also demonstrated acceptable convergent validity with teacher-reported conflict (-0.50 to -0.53). However, student-reported support demonstrated low convergent validity as it was minimally correlated with teacher-reported support ($r = 0.07$ to 0.24) and with student-reported conflict ($r = -0.06$ to -0.14). Consequently, student-reported support was dropped from further analyses.

5.3.5. Academic achievement

Students' academic achievement was assessed using the Woodcock-Johnson III Tests of Achievement (Woodcock et al., 2001) in Grade 3. For students who demonstrated higher Spanish language proficiency, their achievement was assessed using the Bateria III (Woodcock et al., 2005), which yielded scores that were comparable to the WJ-III (Woodcock et al., 2005).

A total achievement score was calculated by creating a composite score of the Broad Math W Score and the Broad Reading W Score. Five subtests were administered to obtain the students' math (i.e., Calculations and Math Fluency) and reading (i.e., Letter-Word Identification, Passage Comprehension, and Reading Fluency) scores. The math subtests were used to measure students' ability to solve arithmetic problems (i.e., Calculation) and to retrieve math facts quickly (i.e., Math Fluency). The reading subtests were used to measure students' ability to identify and read letters and words out loud (i.e., Letter-Word Identification), to understand a sentence or passage by providing the correct missing word (i.e., Passage Comprehension), and to read simple sentences quickly and correctly (i.e., Reading Fluency). Reported internal reliability for both the Broad Math and Broad Reading Clusters were consistently above 0.90 (Woodcock et al., 2001).

5.4. Data analytic strategy

SPSS Version 28 (IBM Corp, 2021) was used to obtain the descriptive statistics and correlations among the main study variables. Mplus Version 8.7 (Muthén & Muthén, 1998–2021) was used to test measurement invariance and to conduct the longitudinal cross-lagged model to address the main research questions.

To assess longitudinal measurement invariance for externalizing behaviors and teacher-student relationships from Grades 3–5, four levels of measurement invariance were tested. Configural invariance examined whether the factor structure was the same over time. Metric invariance examined whether the factor loadings were the same across time. Scalar invariance examined whether the item intercepts were the same across time. Finally, strict invariance constrained factor variance and error variance across time points (Bialosiewicz et al., 2013). The measurement models were compared by examining changes in the comparative fit index (CFI), with a decrease of 0.01 indicating worsening of model fit by imposing an additional level of invariance (Cheung & Rensvold, 2002).

The externalizing behaviors measurement model contained two indicators at each time point (i.e., parent reported hyperactivity and conduct problems) and the teacher-student relationship measurement model contained three indicators (i.e., teacher reported support, teacher reported conflict, and student reported conflict) at each time point. Additional correlations between residuals of the same indicator across time (e.g., correlations between residuals of parent reported hyperactivity in Grades 3, 4, and 5) were added to account for reporter-specific and scale-specific covariance beyond correlations at the factor level.

Upon establishing measurement invariance, we conducted a cross-lagged model to examine the longitudinal relations among student externalizing behaviors, teacher-student relationships, and student classroom engagement (Fig. 1). The cross-lagged model contained the latent constructs of students' externalizing behaviors and teacher-student relationships, as well as the manifest variable

of students' classroom engagement in Grades 3–5. In this model, we tested the longitudinal reciprocal relations between student externalizing behaviors and teacher-student relationships (Hypothesis 1). Additionally, we tested the longitudinal direct and indirect (via teacher-student relationships) predictive paths from externalizing behaviors to classroom engagement (Hypothesis 2). We also assessed the direct and indirect (via externalizing behaviors) effects of teacher-student relationships on classroom engagement (Hypothesis 3). Students' academic achievement and sex were included as covariates in this model. Nonsignificant paths from covariates to main constructs were removed to achieve the most parsimonious model. CFI and root mean square error of approximation (RMSEA) were used to assess model fit. According to [Hu and Bentler \(1999\)](#), a CFI > 0.90 and a RMSEA < 0.08 indicate an acceptable fit.

[Jamshidian et al.'s \(2014\)](#) R package was used to assess the assumptions of multivariate normality and data missing mechanism. The assumption of missing completely at random was met ($p = .35$); however, the data were not multivariate normal. Therefore, the maximum likelihood estimator with robust standard error (MLR) was used to estimate all parameters for the measurement models. Additionally, to assess indirect effects in the longitudinal cross-lagged model, maximum likelihood estimator with 95% bias corrected bootstrap confidence interval based on 5000 bootstrap samples were presented for all parameter estimates. Additionally, given that students were nested within classrooms, the cluster effect at the classroom level was accounted for in all models.

Finally, the sample included students who were native English speakers as well as students who used English as a second language. To ensure that the results were not an artifact of this unique sample composition, we conducted additional analyses using a subsample of students who demonstrated higher English language proficiency at all time points. Overall, results were unchanged from the results based on the entire sample. We reported these additional results in the Supplementary Materials.

6. Results

6.1. Descriptive statistics and correlations

Descriptive statistics are presented in [Table 1](#). On average, parents reported low levels of conduct problems and modest levels of hyperactivity over time. The average levels of conduct problems and hyperactivity in the sample aligned with reports from national norms across several countries ([Becker et al., 2018](#); [Moriwaki & Kamio, 2014](#); [Seward et al., 2018](#)). Teachers reported moderate levels of support, whereas teachers and students reported modest levels of conflict. Teachers reported moderate levels of student classroom engagement.

[Table 2](#) presents correlations among the study variables. All main study variables had modest to moderate levels of stability ($r = 0.66$ – 0.76 for externalizing behaviors; $r = 0.32$ – 0.61 for teacher-student relationships; $r = 0.52$ – 0.55 for classroom engagement). Student conduct problems and hyperactivity were modestly negatively correlated with teacher support ($r = -0.28$ to -0.15) and modestly positively correlated with teacher conflict ($r = 0.22$ – 0.42). Student classroom engagement was correlated moderately positively with teacher support ($r = 0.63$ – 0.65) and negatively with teacher-student conflict ($r = -0.62$ to -0.28). Finally, student conduct problems and hyperactivity were both moderately negatively correlated with classroom engagement ($r = -0.42$ to -0.28).

6.2. Measurement invariance

Model fit indices and model comparisons between different levels of measurement invariance are shown in [Table 3](#). The best fitting and most parsimonious model was the strict error invariance model for both externalizing behaviors, $\chi^2 (df) = 14.054 (13)$, CFI = 0.999, RMSEA = 0.012, and teacher-student relationships, $\chi^2 (df) = 48.622 (28)$, CFI = 0.982, RMSEA = 0.032. Standardized factor loadings from the strict error invariance models are shown in [Table 4](#). All indicators loaded well on the latent factors for both

Table 1
Descriptive Statistics for the Main Study Variables.

	<i>N</i>	% missing	<i>M</i>	<i>SD</i>	<i>Median</i>	<i>Min</i>	<i>Max</i>	Possible Range
Student Conduct Problem (P) G3	477	39%	0.36	0.39	0.20	0.00	1.60	0–2
Student Conduct Problem (P) G4	446	43%	0.35	0.39	0.20	0.00	2.00	0–2
Student Conduct Problem (P) G5	432	45%	0.34	0.37	0.20	0.00	1.80	0–2
Student Hyperactivity (P) G3	477	39%	0.82	0.52	0.80	0.00	2.00	0–2
Student Hyperactivity (P) G4	446	43%	0.79	0.52	0.80	0.00	2.00	0–2
Student Hyperactivity (P) G5	432	45%	0.78	0.51	0.80	0.00	2.00	0–2
Teacher Support (T) G3	547	30%	3.65	0.80	3.75	1.13	5.00	1–5
Teacher Support (T) G4	528	33%	3.60	0.83	3.69	1.13	5.00	1–5
Teacher Support (T) G5	541	31%	3.54	0.82	3.63	1.00	5.00	1–5
Teacher-Student Conflict (T) G3	547	30%	1.79	0.95	1.33	1.00	5.00	1–5
Teacher-Student Conflict (T) G4	528	33%	1.74	0.91	1.33	1.00	5.00	1–5
Teacher-Student Conflict (T) G5	541	31%	1.73	0.92	1.33	1.00	5.00	1–5
Teacher-Student Conflict (S) G3	673	14%	1.77	0.77	1.50	1.00	5.00	1–5
Teacher-Student Conflict (S) G4	670	15%	1.77	0.71	1.67	1.00	5.00	1–5
Teacher-Student Conflict (S) G5	645	18%	1.81	0.74	1.67	1.00	5.00	1–5
Classroom Engagement (T) G3	547	30%	2.82	0.69	2.82	1.09	4.00	1–4
Classroom Engagement (T) G4	528	33%	2.81	0.72	2.82	1.00	4.00	1–4
Classroom Engagement (T) G5	542	31%	2.79	0.73	2.82	1.18	4.00	1–4

Note. *N* = 784 at enrollment. P = parent report; T = teacher report; S = student report; G3–G5 = Grades 3–5, respectively.

Table 2

Correlations between the Main Study Variables.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. Student Conduct Problem (P) G3	–																
2. Student Conduct Problem (P) G4	0.66	–															
3. Student Conduct Problem (P) G5	0.65	0.66	–														
4. Student Hyperactivity (P) G3	0.54	0.43	0.44	–													
5. Student Hyperactivity (P) G4	0.39	0.50	0.45	0.72	–												
6. Student Hyperactivity (P) G5	0.42	0.42	0.54	0.71	0.76	–											
7. Teacher Support (T) G3	–0.25	–0.23	–0.23	–0.26	–0.31	–0.28	–										
8. Teacher Support (T) G4	–0.26	–0.19	–0.31	–0.32	–0.28	–0.34	0.32	–									
9. Teacher Support (T) G5	–0.21	–0.08	–0.15	–0.19	–0.18	–0.25	0.37	0.33	–								
10. Teacher-Student Conflict (T) G3	0.42	0.39	0.36	0.39	0.40	0.40	–0.50	–0.40	–0.30	–							
11. Teacher-Student Conflict (T) G4	0.37	0.36	0.35	0.35	0.34	0.34	–0.40	–0.53	–0.23	0.61	–						
12. Teacher-Student Conflict (T) G5	0.36	0.28	0.30	0.30	0.34	0.35	–0.38	–0.41	–0.50	0.55	0.58	–					
13. Teacher-Student Conflict (S) G3	0.26	0.26	0.25	0.31	0.26	0.31	–0.24	–0.22	–0.18	0.42	0.36	0.33	–				
14. Teacher-Student Conflict (S) G4	0.24	0.24	0.17	0.34	0.30	0.26	–0.22	–0.27	–0.15	0.33	0.39	0.38	0.46	–			
15. Teacher-Student Conflict (S) G5	0.18	0.15	0.23	0.21	0.17	0.22	–0.19	–0.14	–0.24	0.37	0.28	0.42	0.39	0.42	–		
16. Classroom Engagement (T) G3	–0.32	–0.27	–0.27	–0.39	–0.40	–0.38	0.63	0.37	0.35	–0.59	–0.44	–0.40	–0.28	–0.24	–0.22	–	
17. Classroom Engagement (T) G4	–0.33	–0.28	–0.34	–0.39	–0.38	–0.41	0.37	0.65	0.33	–0.43	–0.55	–0.40	–0.28	–0.34	–0.23	0.52	–
18. Classroom Engagement (T) G5	–0.33	–0.26	–0.29	–0.32	–0.36	–0.42	0.45	0.40	0.65	–0.48	–0.42	–0.62	–0.32	–0.30	–0.33	0.55	0.55

Note. P = parent report; T = teacher report; S = student report; G3–G5 = Grades 3–5, respectively. All correlations were significant under the Type I error of 0.05, except for the correlation between parent reported student conduct problems in Grade 4 and teacher reported support in Grade 5.

Table 3
Measurement Invariance: Model Fit and Model Comparison.

Variable	Level of Invariance	χ^2	df	CFI	Δ CFI	RMSEA
Externalizing Behaviors	Configural invariance	4.883	3	0.998	–	0.034
	Metric invariance	7.914	5	0.997	0.001	0.032
	Scalar invariance	8.688	7	0.998	–0.001	0.021
	Strict (factor) invariance	11.194	9	0.998	0.000	0.021
	Strict (error) invariance	14.054	13	0.999	–0.001	0.012
	Configural invariance	21.394	12	0.992	–	0.033
Teacher-Student Relationships	Metric invariance	27.248	16	0.990	0.002	0.031
	Scalar invariance	41.582	20	0.982	0.008	0.039
	Strict (factor) invariance	45.109	22	0.980	0.002	0.038
	Strict (error) invariance	48.622	28	0.982	–0.002	0.032

Note. χ^2 = chi-square; df = degrees of freedom; CFI = comparative fit index; RMSEA = root mean square error of approximation. The best fitting and most parsimonious models are in bold.

externalizing behaviors (standardized factor loadings ranged from 0.57 to 0.92) and teacher-student relationships (standardized factor loadings ranged from –0.84 to 0.49).

6.3. Cross-lagged model

After establishing measurement invariance, a longitudinal cross-lagged model was used to investigate the longitudinal direct and indirect (i.e., mediated by externalizing behaviors) associations between teacher-student relationships and classroom engagement, as well as the longitudinal direct and indirect (i.e., mediated by teacher-student relationships) associations between externalizing behaviors and classroom engagement. For this model, χ^2 (df) = 425.931 (151), CFI = 0.928, RMSEA = 0.052.

Regarding the effects of the two covariates (i.e., students' academic achievement and their sex; Table 5), academic achievement in Grade 3 positively predicted teacher-student relationships in Grade 3, $\beta = 0.23$, 95% CI = [0.14, 0.34], classroom engagement in Grade 3, $\beta = 0.26$, 95% CI = [0.18, 0.34], and classroom engagement in Grade 4, $\beta = 0.12$, 95% CI = [0.03, 0.18]. This suggests that students with better academic achievement were more likely to have a more positive teacher-student relationship and were more engaged in class. Academic achievement negatively predicted students' externalizing behaviors in Grade 3, $\beta = -0.28$, 95% CI = [–0.41, –0.14], suggesting that students who had better academic achievement displayed fewer externalizing behaviors. Additionally, female students had more positive teacher-student relationships, $\beta = -0.35$, 95% CI = [–0.46, –0.23], and higher classroom engagement, $\beta = -0.21$, 95% CI = [–0.30, –0.13], in Grade 3 compared to male students. Male students displayed more externalizing behaviors compared to female students in Grade 3, $\beta = 0.39$, 95% CI = [0.27, 0.51], and Grade 5, $\beta = 0.11$, 95% CI = [0.03, 0.20].

Estimates of autoregressive and cross-lagged effects and their 95% bias corrected bootstrap confidence intervals are presented in Fig. 2. Regarding Hypothesis 1, a more positive teacher-student relationship significantly predicted fewer externalizing behaviors in the subsequent academic year from Grade 3 to Grade 4, $\beta = -0.31$, 95% CI = [–0.57, –0.10], and from Grade 4 to Grade 5, $\beta = -0.23$, 95% CI = [–0.36, –0.09]. Reciprocally, fewer externalizing behaviors also predicted a more positive teacher-student relationship in the subsequent year from Grade 3 to Grade 4, $\beta = -0.24$, 95% CI = [–0.47, –0.05], and from Grade 4 to Grade 5, $\beta = -0.23$, 95% CI = [–0.39, –0.07]. Regarding Hypothesis 2, externalizing behaviors directly negatively predicted classroom engagement in the subsequent year from Grade 3 to Grade 4, $\beta = -0.24$, 95% CI = [–0.54, –0.003], and from Grade 4 to Grade 5, $\beta = -0.31$, 95% CI = [–0.85, –0.13]. However, externalizing behaviors did not indirectly predict classroom engagement through teacher-student relationships, $\beta = -0.04$, 95% CI = [–0.11, 0.42]. Regarding Hypothesis 3, teacher-student relationships did not directly predict classroom engagement from Grade 3 to Grade 4, $\beta = 0.05$, 95% CI = [–0.43, 0.45], or from Grade 4 to Grade 5, $\beta = 0.15$, 95% CI = [–1.07, 0.48]. However, teacher-student relationships indirectly predicted classroom engagement by way of externalizing behaviors, $\beta = 0.10$, 95% CI = [0.01, 0.35]. Specifically, a more positive teacher-student relationship predicted lower subsequent student externalizing behaviors, which in turn predicted higher subsequent classroom engagement.

Table 4
Strict Error Invariance Measurement Model: Standardized Factor Loadings and Standard Errors.

Latent variables	Indicators	Factor loading	S.E.
Externalizing Behaviors	Conduct Problem (P)	0.92*	0.02
	Hyperactivity (P)	0.57*	0.03
	Teacher Support (T)	0.49*	0.04
Teacher-Student Relationships	Teacher-Student Conflict (T)	–0.84*	0.05
	Teacher-Student Conflict (S)	–0.49*	0.04

Note. P = parent report; T = teacher report; S = indicates student report. * = statistical significance under Type I error of 0.05.

Table 5

Cross-Lagged Model: Standardized Path Estimates and 95% Bias Corrected Bootstrap Confidence Intervals.

	Lower 2.5% bound	Estimate	Upper 2.5% bound
Effects of Covariates			
Academic Achievement G3 → Externalizing Behaviors G3	−0.41	−0.28*	−0.14
Academic Achievement G3 → Externalizing Behaviors G4	−	−	−
Academic Achievement G3 → Externalizing Behaviors G5	−	−	−
Academic Achievement G3 → Teacher-Student Relationships G3	0.14	0.23*	0.34
Academic Achievement G3 → Teacher-Student Relationships G4	−	−	−
Academic Achievement G3 → Teacher-Student Relationships G5	−	−	−
Academic Achievement G3 → Classroom Engagement G3	0.18	0.26*	0.34
Academic Achievement G3 → Classroom Engagement G4	0.03	0.12*	0.18
Academic Achievement G3 → Classroom Engagement G5	−	−	−
Sex (0 = female; 1 = male) → Externalizing Behaviors G3	0.27	0.39*	0.51
Sex (0 = female; 1 = male) → Externalizing Behaviors G4	−	−	−
Sex (0 = female; 1 = male) → Externalizing Behaviors G5	0.03	0.11*	0.20
Sex (0 = female; 1 = male) → Teacher-Student Relationships G3	−0.46	−0.35*	−0.23
Sex (0 = female; 1 = male) → Teacher-Student Relationships G4	−	−	−
Sex (0 = female; 1 = male) → Teacher-Student Relationships G5	−	−	−
Sex (0 = female; 1 = male) → Classroom Engagement G3	−0.30	−0.21*	−0.13
Sex (0 = female; 1 = male) → Classroom Engagement G4	−	−	−
Sex (0 = female; 1 = male) → Classroom Engagement G5	−	−	−
Concurrent Correlations			
Teacher-Student Relationships G3 ↔ Externalizing Behaviors G3	−0.80	−0.59*	−0.42
Externalizing Behaviors G3 ↔ Classroom Engagement G3	−0.57	−0.43*	−0.29
Teacher-Student Relationships G3 ↔ Classroom Engagement G3	0.70	0.78*	0.89
Teacher-Student Relationships G4 ↔ Externalizing Behaviors G4	−0.56	−0.10	0.15
Externalizing Behaviors G4 ↔ Classroom Engagement G4	−0.28	−0.10	0.08
Teacher-Student Relationships G4 ↔ Classroom Engagement G4	0.67	0.75*	0.93
Teacher-Student Relationships G5 ↔ Externalizing Behaviors G5	−0.51	−0.25*	−0.03
Externalizing Behaviors G5 ↔ Classroom Engagement G5	−0.54	−0.36*	−0.20
Teacher-Student Relationships G5 ↔ Classroom Engagement G5	0.70	0.78*	0.92

Note. G3–G5 = Grades 3–5, respectively; – = nonsignificant covariate effects that were dropped from the final model. * = statistical significance under Type I error of 0.05.

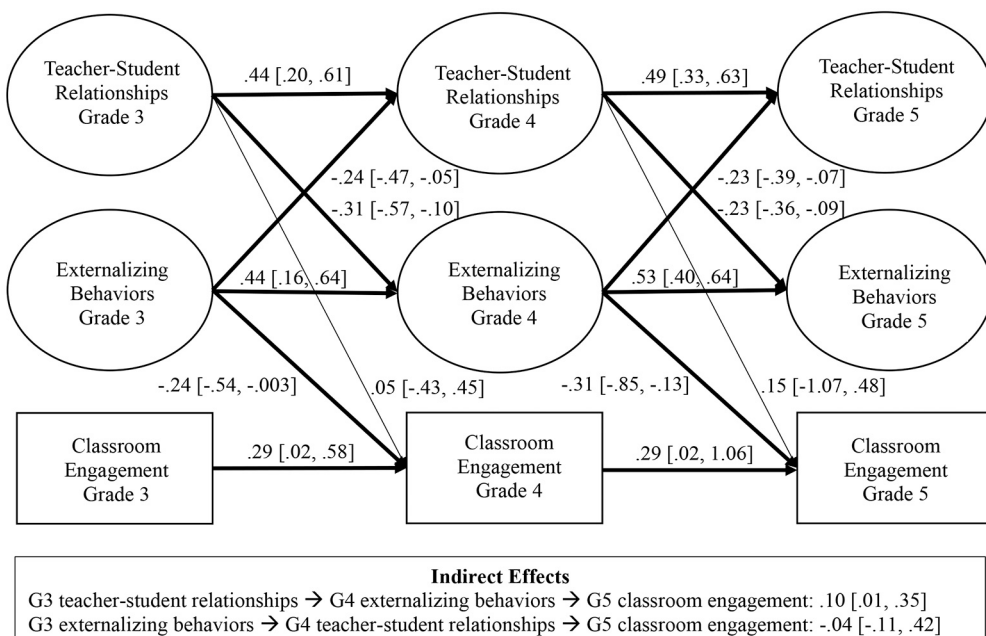
Parameter Estimates and 95% Bias Corrected Bootstrap Confidence Intervals for Autoregressive and Cross-lagged Paths

Fig. 2. Parameter Estimates and 95% Bias Corrected Bootstrap Confidence Intervals for Autoregressive and Cross-lagged Paths.
 Note. Effects of covariates and concurrent correlations are not shown for simplicity of presentation. Significant paths are bolded.

7. Discussion

Taking a person-environment transactional perspective, the present study examined how students' classroom engagement developed as a function of the dynamic interaction between students and their social relational contexts. This work addresses an important gap in the literature that has historically investigated the roles of personal and contextual influences on engagement behaviors separately. In the present study, we examined the joint longitudinal direct and indirect effects of student externalizing behaviors and teacher-student relationships on the development of students' classroom engagement by using a multi-informant approach in a 3-year longitudinal design.

Our first hypothesis of a negative reciprocal relation between students' externalizing behaviors and teacher-student relationships was supported. Specifically, parents reported children having fewer externalizing behaviors when teachers and students reported having a more supportive and less conflictual teacher-student relationship during the previous year. Similarly, teachers and students reported a more positive relationship when parents reported fewer externalizing behaviors the previous year. This finding is consistent with several longitudinal studies that examined the role of a positive teacher-student relationship in mitigating students' development of externalizing behaviors, as well as the role of externalizing behaviors in escalating tensions and conflicts in the teacher-student relationship (Crockett et al., 2018; Doumen et al., 2008; Harvey et al., 2022; MacLean et al., 2023; O'Connor et al., 2011; Silver et al., 2005; Williford et al., 2017; Zhang & Sun, 2011). This finding is aligned with the person-environment transactional perspective (Sameroff, 2009; Scarr & McCartney, 1983) that highlights the reciprocal process in which the behavioral development of an individual not only *depends* on their social relationships, but also *brings* changes in their social relationships.

The second hypothesis (i.e., externalizing behaviors negatively predict subsequent student classroom engagement both directly and indirectly through teacher-student relationships) was partially supported. Specifically, externalizing behaviors predicted subsequent classroom engagement directly. This finding of a negative longitudinal association between externalizing behaviors and classroom engagement is consistent with several existing cross-sectional (Fantuzzo et al., 2005; Olivier, Morin, et al., 2020; Tian et al., 2017) and longitudinal studies (Searle et al., 2013), demonstrating the elevated behavioral and academic risks among students with high externalizing behaviors. In addition, our unique focus on the longitudinal indirect effects of externalizing behaviors on classroom engagement offers a unique opportunity to investigate whether children's externalizing behaviors negatively predict their classroom engagement by way of their social relationships, a question that has not yet been addressed by the literature. Our findings did not support such an indirect longitudinal effect, suggesting that an increasingly negative teacher-student relationship and the correspondingly decreasing social motivations are unlikely the primary mechanism for the low engagement among students with high externalizing behaviors. Instead, the finding of a direct predictive effect of externalizing behaviors on the development of classroom engagement suggests that students with more externalizing behaviors experience more difficulties in regulating attention, emotion, and behaviors (Rothbart & Bates, 2006), which may directly limit their capacity to engage in effective learning in the classroom setting that involves following classroom rules, following instructions, and sustaining attention on tasks (Baker et al., 2008; Campbell et al., 2014). Our novel approach to take the person-environment transactional perspective provides us with these critical insights into the possible and unlikely mechanisms for the development of classroom engagement among students with elevated externalizing behaviors, which significantly contributes to the existing literature that has traditionally investigated the roles of personal and contextual influences on classroom engagement separately.

Our finding of a lack of indirect effect of externalizing behaviors on students' classroom engagement via teacher-student relationship contrasts with a recent cross-sectional study that found that a sense of relatedness to school mediated the links between behavior and social problems and classroom engagement (Olivier, Archambault, & Dupere, 2020). This discrepancy may be attributable to the different types of personal attributes being examined in these two studies, with the present study focusing exclusively on externalizing behaviors whereas the Olivier, Archambault, and Dupere study focused on co-occurring externalizing, internalizing, and social problems. As social problems are conceptually more closely related to teacher-student relationship quality and sense of relatedness, a combined behavioral and social problem construct may exhibit a stronger indirect predictive effect on students' classroom engagement via relational measures. It is also possible that the current use of a longitudinal design is better equipped than the cross-sectional design in Olivier, Archambault, and Dupere to elucidate the temporal precedence in the associations among variables. Future studies would benefit from examining whether teacher-student relationships mediate the associations between classroom engagement and other types of behavioral problems such as internalizing and social problems, in longitudinal designs.

Our final hypothesis (i.e., teacher-student relationships positively predict subsequent student classroom engagement both directly and indirectly through student externalizing behaviors) was partially supported. Specifically, teacher-student relationships only indirectly predicted subsequent classroom engagement through externalizing behaviors. This finding highlights the unique contribution of the present study to the existing literature that has primarily conceptualized teacher-student relationships as the context of students' classroom engagement without considering the active role of students in their dynamically evolving social relational context (Furrer et al., 2014; Klem & Connell, 2004; Quin, 2017; Roorda et al., 2017). Our finding critically adds to the existing literature in two ways. First, teacher-student relationships did not uniquely directly predict the development of classroom engagement once students' levels of externalizing behaviors were controlled for. This result is significantly different from previous findings of significant predictive effects of teacher-student relationships on classroom engagement when students' personal attributes are not considered simultaneously (e.g., Furrer et al., 2014; Klem & Connell, 2004; Quin, 2017; Roorda et al., 2017). Such a difference is critical as it highlights the necessity of taking into consideration students' personal attributes when identifying contextual influences that uniquely support or undermine classroom engagement above and beyond students' pre-existing individual differences. Second, although teacher-student relationships did not directly predict classroom engagement, it did predict classroom engagement indirectly through externalizing behaviors. This finding is novel and important as it highlights a mechanism through which relational factors in the

classroom may predict students' learning behaviors. Specifically, teacher-student relationship quality is important in predicting students' classroom engagement primarily to the extent that it may reinforce those students' characteristics (e.g., externalizing symptoms) that limit students' capacity to effectively focus and participate in the classroom.

In the only other study that simultaneously investigated externalizing behaviors and teacher-student relationships in relation to classroom engagement, Archambault et al. (2017) reported that a positive teacher-student relationship protects elementary school students with high levels of oppositional behaviors from disengaging in the classroom. Taking a person-environment transactional perspective, the present study critically complements this finding by highlighting the non-independent and reciprocal nature of the association between students' externalizing behaviors and teacher-student relationships as well as its longitudinal predictive effects on students' classroom engagement. Our findings revealed that not every student has an equal opportunity to enjoy a positive teacher-student relationship; students with more externalizing behaviors tend to receive less support from and experience more conflict with their teachers. This is particularly salient when a teacher who is highly warm and sensitive to one student may react less supportively and more harshly to another student who has more externalizing behaviors. This negative teacher-student relationship may then reinforce students' externalizing behaviors, which further contributes to compromising these students' capacity to fully engage in classroom learning. These processes together form a cycle that makes it extremely challenging for students with high externalizing behaviors to thrive in the classroom. These findings suggest that simply placing students with externalizing behaviors with a supposedly warm and sensitive teacher will unlikely improve their classroom engagement in the long run. To generate lasting effects, practical efforts should acknowledge and address not only issues presented among students with externalizing behaviors, but also the possible long-term risks among teachers who regularly work with these students, such as elevated levels of stress and negative reactivity toward specific students (e.g., those with high externalizing behaviors; Alamos et al., 2022; Baeriswyl et al., 2021; Camacho et al., 2022).

8. Strengths and limitations

The present study is characterized by a few notable strengths. First, this study utilized a multi-wave longitudinal design that allowed for the investigation of the dynamic transactions among students' externalizing behaviors, teacher-student relationships, and students' classroom engagement over the course of 3 elementary school years. Additionally, this study implemented a multi-informant approach by gathering information from the perspectives of parents, teachers, and students. This measurement approach minimizes shared variance among variables due to measurement artifacts and boosts our confidence over the substantive meaning of the findings.

One noteworthy limitation of the present study is that only parent report was used to measure externalizing behaviors. Parental reports of externalizing behaviors are context-specific, which may not fully capture the extent to which students exhibit similar behaviors in the classroom setting. That said, Stone et al. (2010) reported moderate levels of inter-rater agreement between parent and teacher reported hyperactivity and conduct problems using the SDQ, indicating common variance in parent and teacher reported externalizing behaviors across contexts. Given that teacher reports were used for all other main constructs in the present study, using parent reports for externalizing behaviors brought the benefits of minimizing reporter-specific method artifact, which offered a conservative but substantively meaningful estimate of the direct and indirect effects. Relatedly, teacher report was used to measure both students' classroom engagement and teacher-student relationships. Teachers' perception of student behaviors and teacher-student relationships may be biased based on students' demographic characteristics (Rudasill et al., 2023; Wymer et al., 2022). For example, racial and ethnically minoritized students, such as Black students, are often perceived by their teachers to be less motivated, less academically skilled, and more disruptive in the classroom than White students (Brandmiller et al., 2020; Cherng, 2017; Rudasill et al., 2023; Santiago-Rosario et al., 2021; Wymer et al., 2022). Such disparities in the perception of White versus racial/ethnically minoritized students are particularly salient among White teachers (Wymer et al., 2022). Such teacher perceptual biases may play a role in accounting for the present findings given our racial/ethnically diverse student sample and predominantly White teacher sample (i.e., over 80% White teachers). Future research should examine the replicability of the present findings by gathering additional information from multiple perspectives (e.g., direct classroom observations and student self-report) to measure all key constructs.

Additionally, the present study utilized a correlational design. Thus, we were not able to make definitive causal inferences regarding the associations among students' externalizing behaviors, teacher-student relationships, and classroom engagement behaviors as other confounding variables that were not included in the present study may contribute to explain the observed associations. Finally, data were collected two decades ago, which may limit their generalizability to more recent cohorts. Future research based on more recent data are needed to investigate the replicability of the current longitudinal mediation findings.

9. Conclusions and implications

The present study highlighted the longitudinal predictive roles of both personal characteristics (e.g., externalizing behaviors) and relationship qualities (e.g., teacher-student relationships) in the development of classroom engagement. Although the data were collected two decades ago, our findings may assist in understanding the critical mechanisms underlying the low levels of classroom engagement among students with high externalizing behaviors. Although the mean level of classroom engagement and the prevalence of externalizing behaviors may shift over time, the mechanisms underlying their developmental relations are likely consistent across cohorts. As such, this theoretical level understanding of the development of classroom engagement situated within a person-environment transactional framework has timely and important practical implications for contemporary education practices, especially considering the increasing national rates of classroom disruptive behaviors, teachers' sense of unpreparedness, and lack of support in classroom management (EAB, 2019).

Our findings showed that a student's disengagement in the classroom may arise from their externalizing tendency and reinforced by a poor relationship with their teachers. Without timely prevention, these transactional processes will possibly cascade into a constellation of negative outcomes in the behavioral, social-relational, and academic domains for some students, who may ultimately require more intensive and costly treatments and rehabilitation. As such, brief universal prevention programs (e.g., Establish-Maintain-Restore [EMR] method; Cook et al., 2018) that target the general student population with subclinical levels of externalizing behaviors may prove efficient and cost-effective to engage a larger population of students in the classroom. Although the present findings may not generalize to a clinical sample, these clinical implications are highly relevant for school psychologists as they consider distributing time and resources in prevention efforts at the general student population level.

In addition, given that negative teacher-student relationships may serve to reinforce students' externalizing behaviors and contribute indirectly to the development of disengagement in the classroom, awareness needs to be raised regarding how teachers' own behaviors may impact the trajectory of students' behavioral development. The ways in which teachers respond to students' externalizing behaviors may model acceptable behaviors in social interactions. A patient, warm, and supportive response from the teacher models a nonreactive and well-regulated way of deescalating conflict, which may be key to mitigating further development of externalizing behaviors (Baker et al., 2008; Harvey et al., 2022; Leflot et al., 2011; MacLean et al., 2023; O'Connor et al., 2011; Silver et al., 2005; Williford et al., 2017). Yet, teachers do not typically receive extensive training or support for the development of social emotional skills to intentionally create and maintain supportive and nonreactive teacher-student relationships across diverse student populations. School psychologists can facilitate the implementation of universal prevention practices, such as the EMR method, to equip teachers with positive relationship building and maintenance skills (Cook et al., 2018; Kincade et al., 2020).

Author notes

L.M.H. and Z.W. conducted the analyses. L.M.H., M.Q., and Z.W. drafted the manuscript. All authors provided constructive feedback on the development of this manuscript.

Declaration of Competing Interest

None.

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Appendix A. Supplementary data

Supplementary materials for this article can be found online at <https://doi.org/10.1016/j.jsp.2023.101242>.

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